

Algorithm 2: E commerce Platform Search Function

Exercise 2: E-commerce Platform Search Function

Scenario:

You are working on the search functionality of an e-commerce platform. The search needs to be optimized for fast performance.

Steps:

Understand Asymptotic Notation:

Explain Big O notation and how it helps in analyzing algorithms.

Describe the best, average, and worst-case scenarios for search operations.

Setup:

Create a class Product with attributes for searching, such as productId, productName, and category.

Implementation:

Implement linear search and binary search algorithms.

Store products in an array for linear search and a sorted array for binary search.

Analysis:

Compare the time complexity of linear and binary search algorithms.

Discuss which algorithm is more suitable for your platform and why.